Week 04 — IoT Connectivity Technologies

> Slides: week04-iot-edge-04.pdf

Learning objectives: identify key connectivity requirements, explain wireless communication basics, compare major IoT connectivity technologies, explain network topologies and gateways, and **select suitable connectivity for an IoT application**.

The selection chain used all lecture:

```
Application requirements -> Connectivity requirements -> Physical communication constraints
 -> Candidate technologies -> Technology comparison -> Connectivity selection
```

Contents: 1. Connectivity fundamentals · 2. Local and short-range technologies · 3. Wide-area connectivity and technology selection

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Part 1 — IoT Connectivity Fundamentals

Connection from week 3

Week 3 — what the device sends	**Week 4 — how the device establishes a link**
Physical quantity → sensor → IoT device → processing → structured data → **serialized format**	**Serialized format** → communication interface → **connectivity technology** → receiver / gateway / network

What is IoT connectivity?

> **IoT connectivity** = a **wired or wireless communication link** that allows an IoT device to **exchange data** with another device, gateway, access point, base station or network.

Examples of connectivity paths:

- Wearable — **Bluetooth** → smartphone
- Camera — **Wi-Fi** → access point
- Machine — **Ethernet** → industrial network
- Farm sensor — **LoRa** → gateway
- Tracker — **cellular** → base station

Connectivity vs communication protocol

Connectivity	**Communication protocol**
How the devices get connected	**How data is exchanged once the connection exists**
Wi-Fi, BLE, Ethernet, LoRa, cellular	Covered in **week 5**

```
IoT device -> Connectivity technology -> Communication link established -> Communication protocol -> Application / server
```

> **Analogy: connectivity technology = the road or highway; communication protocol = the rules for how information travels on that road.**

Wi-Fi gives you a wireless link, but it does **not** define how an application exchanges sensor messages with a server. The protocol answers: where is the destination, how should data be delivered, what happens if communication fails, and how should an application request or publish information.

Why not use Wi-Fi for everything?

Application	Main characteristics	Suitable technology
Security camera	High data volume, continuous video, fixed location, power available, local network	**Ethernet / Wi-Fi**
Wearable health sensor	Small data, battery powered, worn on body, phone nearby	**BLE**
Farm temperature sensor	Low data rate (~10 bytes, e.g. `{"temperature": 24.3, "battery": 81}`), long range, battery/solar	**LoRa**
Delivery vehicle tracker	Moving, city-wide/regional, moderate or low data, no fixed nearby gateway	**LTE / cellular**

> **Technically possible ≠ architecturally appropriate.**

The method: first ask "*What does this application need?*", then "*Which technology matches those requirements?*"

Common connectivity requirements: range, data rate, power, mobility.

Requirement 1: Range

> **Range** = the **maximum practical communication distance** between communicating nodes.

Scale	Technology
Centimeters	**NFC** — contactless payment, transit passes
Meters (local/short)	**BLE**
Tens/hundreds of meters	**Wi-Fi**
Kilometers	**LoRa**
Wide area	**Cellular**
Regional/global	**Satellite**

> **"Range = 100 m" does NOT mean "always works at 100 m".** Real range depends on transmitter power, antenna, frequency, receiver sensitivity, obstacles, interference and environment. **100 m in an open field ≠ 100 m through factory walls.**

Requirement 2: Data rate

> **Data rate** = amount of digital data transferred per unit time, typically **bit/s**.

- Temperature sensor: **25–100 bytes every 1–10 minutes** — it does not need a 1 Gbit/s link.
- Video camera: **continuous stream**, possibly millions of bits per second. (A 1 Gbit/s link ≈ **125 MB/s**.)

Bandwidth vs data rate: bandwidth is the *maximum* amount transmittable in a given time; **data rate** is the rate actually transferred. Greater bandwidth can support higher data rates, but **the two are not identical**.

Requirement 3: Power consumption

A wireless **transceiver (radio)** operates in states: **transmit, receive, idle/listen, sleep**.

State	Power demand
Transmit / receive	**High** activity
Idle / listening	Moderate
Sleep	**Very low**

> **Design objective for battery devices: keep the radio asleep as much as possible, and activate it only when communication is necessary.**

Example 1 — mains-powered security camera: continuous power, continuous connectivity, high-volume video, frequent communication. Communication energy is **not** the dominant constraint; prioritize **data rate, stable connection, latency, reliability**.

Example 2 — coin-cell wearable: very small battery, no convenient charging, lifetime of months, tiny data (heart rate 78 bpm, temperature 36.7 °C, battery 83 %). The pattern is:

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Sleep for most of the time -> Wake briefly -> Measure sensor -> Transmit a small message -> Return to sleep
```

Why radio activity matters — the calculation ($E \approx I \times t$):

	Device A	**Device B**
Behaviour	Transmit 1 s **every minute**	Transmit 1 s **every hour**
Transmissions/day	60 × 24 = **1440**	**24**
Transmit time/day	1440 s = **0.4 h**	24 s = **0.00667 h**
At 30 mA	30 × 0.4 = **12 mAh/day**	30 × 0.00667 = **0.20 mAh/day**

> **Device A consumes about 60× the transmit energy of device B.**

Real battery consumption is more complicated: energy is also spent on sensor measurement, processor operation, **radio startup**, transmission, reception, listening, waiting, **retransmission** and sleep. Still, **communication frequency and radio-on time strongly influence battery life**.

Duty cycle = the **fraction of time the radio is active** compared with total time.

- Device A: 1 s every 60 s → **1.67 %**
- Device B: 1 s every 3600 s → **0.028 %**

> **Lower duty cycle → less radio-on time → lower communication energy → longer battery life.**

More data also means more radio-on time (20 bytes vs 2 MB): more transmission time → more radio activity → more energy.

The balance: a slowly changing temperature sensor **should not** transmit every second — unnecessary communication consumes battery and network resources. But an **emergency gas-leak sensor** needs **low latency and a rapid alarm**. **Connectivity design always balances multiple requirements.**

Requirement 4: Latency

> **Latency** = delay between sending information and its arrival/use at the destination.

Application	**Delay tolerance**
Soil moisture monitoring	Seconds/minutes acceptable
Building temperature	Seconds often acceptable
Emergency alarm	Much lower delay desired
Time-sensitive control	**Very low delay required**

Requirement 5: Mobility

> **Mobility** = whether a device **changes physical location while operating**, and whether communication must **continue while it moves**.

Mobility level	**Example**	**Communication requirement**
Fixed	Factory machine	Stays in one place → **Ethernet** is fine
Local mobility	Smartwatch	Moves within a limited local area → **BLE** within range of the phone
Wide-area mobility	Scooter / vehicle	Moves over large geographic areas → **cellular**

Rental scooter example: it travels across areas A–D, needs location and battery/status reporting, so the system must stay available across a **large geographic area**. **Wi-Fi is less suitable** — once the scooter leaves coverage, there is no connection.

Range vs mobility:

	Range	Mobility
Main question	**How far can the link reach?**	**Can communication continue as the device moves?**
Example	Sensor 3 km from a gateway	Scooter travelling through a city
Relevant factor	Radio/link distance	**Coverage + infrastructure while moving**

Other selection factors

Factor	What it adds
Device density	Not just one device — **how many devices compete for or share the infrastructure**
Infrastructure	What already exists. A wearable can use the phone's **Bluetooth**; a remote farm sensor with **no Wi-Fi nearby** needs something else (e.g. cellular)
Environment	**Concrete** weakens/absorbs signals; **steel** reflects and blocks; **machinery** causes electromagnetic interference; **underground structures** severely block signals
Cost and reliability	Is occasional loss acceptable? Is availability critical? Is the infrastructure privately controlled? Is **redundant coverage** required?

> **Connectivity selection depends on engineering performance *and* deployment constraints.**

The connectivity trade-off triangle

> A system rarely maximizes **long range**, **high data rate** and **very low power** simultaneously.

Choose based on application requirements, balance the three, and accept that **no single technology is best for every use case**.

Short discussion: four scenarios

Scenario	Range	Data	Power	Mobility	Priority	Choice
A. Security camera	Local building	High — continuous video	Low concern if mains-powered	Fixed	**Data rate**	Ethernet / Wi-Fi
B. Wearable heart-rate patch	Short	Low — small periodic measurements	**Very important** (battery)	Moves with person	**Low power**	BLE
C. Farm soil sensor	Long — several km	Very low — tiny periodic message	**Very important** (battery)	Fixed	**Long range, low power**	LoRa
D. Delivery tracker	Wide area — city/regional	Low to moderate — periodic location	Important	**Highly mobile**	**Mobile, wide-area coverage**	Cellular

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Part 2 — Local and Short-Range Connectivity Technologies

Connectivity categories

Wired	Local / short-range wireless	Identification / very short range	Wide area
Ethernet	Wi-Fi, BLE, IEEE 802.15.4	RFID, NFC	LoRa, NB-IoT, LTE-M, 4G/5G, satellite

Wired vs wireless

Wired (physical cable)	Wireless (radio)
Stable link	**Flexible deployment**
High throughput possible	**Mobility possible**
Less radio-frequency interference	Subject to propagation/interference
Installation needed	**Coverage constraints**

Ethernet

Strengths	Limitations
Very high data rate	**Physical cable required**
Stable, reliable physical link	Installation effort
Low susceptibility to RF interference	**Limited mobility**
Suitable for continuous communication	

Typical IoT roles: industrial machines, gateways, fixed cameras, building equipment.

Power over Ethernet (PoE): delivers **electrical power and network data over the same Ethernet cabling** — **one cable = data + power**. Useful for IP cameras, wireless access points, gateways, and industrial sensors/controllers.

Wi-Fi

Characteristics: local-area, **high throughput**, existing infrastructure is common, and **relatively high power** compared with low-power IoT radios. Best for **local-area devices where Wi-Fi infrastructure already exists**.

Limitations: limited range (especially rural); requires an **access point and power**, which may not be available or economical to deploy; and its **relatively high power drains a coin-cell battery** — wasteful when a sensor only sends a tiny amount of data.

BLE — Bluetooth Low Energy

> **BLE** = low-power communication designed for **battery-powered devices**, **short range** (tens of meters), commonly used for **device-to-phone** or **device-to-gateway** communication, ideal for **sensor data and simple control at low data rates**.

Why BLE is low power: the radio **stays off most of the time**, **wakes briefly** to send or receive small data, and the **short active time reduces energy use** — the duty-cycle principle from Part 1.

Use cases: syncing with a phone (notifications, health data), monitoring vital signs, activity/step/sleep tracking, indoor navigation and asset tracking, and sending sensor readings (temperature, humidity, environmental data) to a smartphone.

Wi-Fi vs BLE

Requirement	Wi-Fi	BLE
Data rate	Higher	Lower
Power	Higher	**Very low**
Typical range	Local	Short/local
Typical infrastructure	**Access point**	**Phone/gateway**
Typical application	Camera	Wearable

IEEE 802.15.4

> Specifies a **low-rate, low-power wireless foundation** used by several IoT networking technologies (e.g. Zigbee).

Key characteristics: low data rate (small, infrequent transmissions such as sensor measurements), **low power** (battery devices), **short to medium range** (tens to hundreds of meters), and suitability for **many sensor nodes** — IoT applications with lots of low-power devices.

Network topology

> **Topology** describes **how communicating nodes are arranged and connected**.

Topology	Arrangement	Advantages	Limitations	Example
Point-to-point	Two endpoints communicate directly	Simple	Only two nodes	Sensor → controller
Star	Multiple devices connect through **one central node**	Simple; easy centralized management	**Central point dependency**; every endpoint must reach the center	Devices → Wi-Fi AP
Mesh	Nodes connect **through neighboring nodes**	**Extended coverage**; **alternative paths**	More complexity (maintaining relationships); **relay burden**	Building sensor network

Mesh relay burden: a node must send **its own data and relay another node's data** → more relay activity → more radio-on time → **more energy consumption**.

RFID — Radio Frequency Identification

> **RFID** uses radio waves to **automatically identify and track tags** attached to objects, animals or people **without requiring a direct line of sight**. **Line of sight (LoS)** = an unobstructed straight path through the air between transmitting and receiving antennas.

Feature	Barcode	QR code	RFID
Reading method	Optical	Optical	**Radio wave**
Line of sight	Required	Required	**Not required**
Multiple items	Usually one at a time	Usually one at a time	**Multiple tags possible**
Cost	Very low	Very low	Higher
Typical use	Product ID	Links/tickets/info	**Inventory/assets**

NFC (near-field communication) is the **centimeter-range** relative, used for contactless payment and transit passes.

Local technology comparison

Technology	Relative range	Data rate	Power	Typical use
Ethernet	Cable	Very high	Usually powered	Machine / gateway / camera
Wi-Fi	Local	High	Medium–high	Camera / appliance
BLE	Short	Low–medium	**Very low**	Wearable
802.15.4 / Zigbee	Short–medium	Low	Low	Sensor network
RFID / NFC	Very short (reader-dependent)	Low	**Very low / passive**	Identification

Short discussion: choose a technology

Case	Technology	Range reason	Data-rate reason	Power reason	Infrastructure reason
Smartwatch → phone	**BLE**	Short-range communication	Small/moderate	Very low power	The smartphone already supports BLE and acts as the nearby central device
Machine-vision camera → control room	**Ethernet or Wi-Fi**	Local scale	Video requires high data rate	Camera is usually mains-powered	Ethernet preferred if cabling exists; Wi-Fi if cabling is difficult and an AP is available
Warehouse pallet identity	**RFID**	Only reader-to-tag range	Only a small identifier needs transferring	**Passive tags may operate without a battery**	Install fixed or handheld RFID readers
Building temperature sensors	**IEEE 802.15.4**	Short-to-medium; multiple nodes can cover a building	Readings are very small and periodic	Low power	A coordinator can collect data from many sensor nodes

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Part 3 — Wide-Area Connectivity and Technology Selection

Local vs wide-area connectivity

Local connectivity	Wide-area connectivity
Short-range connection	Broader-area connectivity
Local access to nearby devices or a local network	Connects devices over long distances, across **cities, regions or globally**

Why local isn't enough: **in rural and remote areas, local connectivity may simply not be available.**

LPWAN — Low-Power Wide-Area Network

> **LPWAN** = wireless technologies designed for IoT that provide **long-range communication with very low power consumption**. **Trade-off: very low data rate.**

Cellular LPWAN (licensed spectrum)	Non-cellular LPWAN (unlicensed spectrum)
Run by **telecom operators**	Run on **open radio bands**
Example: **NB-IoT, LTE-M**	Example: **LoRa**

LPWAN can use little power because the battery benefits from a **low duty cycle**.

LoRa

> **LoRa** = a long-range, low-power **radio modulation / physical-layer** technology. It shapes radio waves so they **travel long distances on very low power**, at **low data rates**.

LoRa vs LoRaWAN:

LoRa	LoRaWAN
The **radio modulation** used to transmit data over the air	The **network protocol**: defines how devices, gateways and network services communicate

In real deployments: **device → LoRa/LoRaWAN → gateway → Ethernet/cellular/Wi-Fi → internet**.

Strengths	Limitations
Long range	**Very low data rate**
Low power	Not for high-bandwidth applications
Private gateway possible	Gateway/network infrastructure still required
Useful in remote sensing	

Cellular IoT

IoT devices use cellular networks to reach the internet **via a mobile operator**: **wide geographic coverage**, **operator-managed infrastructure**, **mobility support**, and **subscription/service dependency**.

LoRa vs cellular:

	LoRa	Cellular
Primary target	Simple machines and industrial sensors	Smartphones, tablets, laptops
What it transmits	**Tiny numbers** (GPS points, temperature)	HD video, websites, apps, voice
Bandwidth	**Extremely low** (a few bytes per message)	**Massive** (gigabytes per second)
Network infrastructure	**Private or public**	**Public only** (must use a carrier's tower)
Battery consumption	**Ultra-low** (can last 10 years on one battery)	High (drains phone batteries in 1–2 days)

NB-IoT and LTE-M

NB-IoT — a **cellular LPWAN** using the existing cellular network, operated by telecom companies: **wide-area coverage, low data rate, low power, good indoor coverage**, suitable for **relatively static, low-data devices** (e.g. **smart meters**).

LTE-M — cellular IoT for **connected, mobile things**: leverages existing **LTE infrastructure** for broad coverage, supports **higher data rates**, **handles moving devices more efficiently**, and is well suited to **small, mobile, battery-powered devices**. (LTE-M is for machines — sensors, trackers; LTE is for smartphones and tablets.)

	NB-IoT	LTE-M	LoRa
Data	Lower	Higher	Very low
Mobility	Limited / lower focus	**Better**	Limited
Power	Low	Low–moderate	Low
Infrastructure	Operator network	Operator network	**Private / local gateway possible**
Typical application	**Meters**	**Trackers**	**Farm / environmental sensors**

Full technology comparison

Technology	Range	Data	Power	Mobility	Dependency
Ethernet	Cable	Very high	Powered	None	Cabling
Wi-Fi	Local	High	Medium–high	Moderate	Access point
BLE	Short	Low–medium	Very low	Good (within range)	Phone / gateway
802.15.4	Short–medium	Low	Low	Limited	Coordinator / gateway
LoRa	Long	Very low	Very low	Limited/moderate	Gateway
NB-IoT	Wide	Low	Low	Limited	Cellular operator
LTE-M	Wide	Low–medium	Low–medium	Good	Cellular operator
4G/5G	Wide	High	Higher	Excellent	Cellular operator
Satellite	Very wide	Low–medium	Higher	Good	Satellite service

> ⚠️ This table's columns are **misaligned in the slide**; the version above is the reconstructed mapping. Check it against the slide if it is used as an exam answer.

Connectivity selection framework — the ten questions

1. How far? 2. How much data? 3. How much power? 4. How quickly? 5. Moving or fixed?
2. How many devices? 7. What infrastructure exists? 8. What environment? 9. What reliability? 10. What cost? → **Select technology.**

Worked cases

Case 1 — Machine vision. An industrial camera inspecting a production line, connected to the control room. *Reasoning:* high data (video needs high bandwidth), fixed installation, power available, reliability (wired is stable). → **Ethernet**, alternative **Wi-Fi** if cabling is difficult.

Case 2 — Wearable. A body-worn sensor communicating with a smartphone. *Reasoning:* short range (meters) is sufficient, small data amounts, low power suits battery operation, smartphone available as a nearby gateway. → **BLE**. *Why not LTE-M?* Greater complexity (modem, SIM) and higher energy and service cost.

Case 4 — Rental scooter. A connected scooter reporting location and status. *Reasoning:* wide coverage across a city, supports mobility, uses existing operator infrastructure. → **LTE-M / cellular**, a good balance of coverage, mobility and power efficiency.

(The slides jump from case 2 to case 4.)

Summary

```
Requirement -> Candidate connectivity technology -> Physical / infrastructure fit -> Trade-off justification
Device -> Connectivity link -> Network / gateway
```

Next: once the connectivity link exists, **how is information addressed, transported and exchanged? → communication protocols (week 5)**.

Most important for exams

- **Connectivity vs protocol:** connectivity = **how devices get connected** (the road); protocol = **how data is exchanged once connected** (the rules of the road). Week 4 vs week 5.

- **Four core requirements: range, data rate, power, mobility** — plus latency, device density, infrastructure, environment, cost and reliability.
- **"Range = 100 m" never means "always works at 100 m"** — power, antenna, frequency, obstacles, interference, environment all matter.
- **Bandwidth ≠ data rate** (maximum possible vs actually transferred).
- **Radio states:** transmit/receive (high), idle/listen (moderate), **sleep (very low)**. Design goal: **keep the radio asleep**.
- **Duty cycle = active time ÷ total time.** Be ready to compute it (1 s per 60 s = **1.67 %**; 1 s per 3600 s = **0.028 %**) and the energy (**E ≈ I × t**: 30 mA × 0.4 h = **12 mAh/day** vs **0.20 mAh/day**, a **60×** difference).
- **Lower duty cycle → longer battery life; more data → more radio-on time → more energy.**
- **Mobility levels:** fixed (factory machine, Ethernet), local (smartwatch, BLE), wide-area (scooter, cellular). **Range vs mobility** = how far the link reaches vs whether it continues while moving.
- **Trade-off triangle:** you cannot maximize **long range + high data rate + very low power** at once.
- **The four canonical pairings:** camera → **Ethernet/Wi-Fi**; wearable → **BLE**; farm sensor → **LoRa**; vehicle tracker → **cellular/LTE-M**.
- **Topologies:** point-to-point, **star** (central dependency), **mesh** (extended coverage and alternative paths, but complexity and **relay burden → more energy**).
- **RFID vs barcode/QR:** radio vs optical, **no line of sight needed**, **multiple tags at once**, higher cost, **passive tags need no battery**. NFC = centimeters.
- **PoE** = data + power over one Ethernet cable.
- **LPWAN** = long range + very low power, **trade-off: very low data rate**. **Cellular LPWAN** (NB-IoT, LTE-M, licensed, operator-run) vs **non-cellular** (LoRa, unlicensed, private gateway possible).
- **LoRa = the radio modulation; LoRaWAN = the network protocol.**
- **NB-IoT vs LTE-M:** meters (static, low data, good indoor) vs trackers (mobile, higher data rate).
- **The ten selection questions**, and the ability to justify a choice with range / data / power / mobility / infrastructure reasons.